

Supplemental Material: Honest Topological Ginzburg–Landau Dynamics

Truncated GPE, live gap, dynamical Casimir ODE, and G_{ij}
Companion to QVC arXiv v4 (Option A*)

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Abstract

This supplement specifies the dynamical content of the Quantum Vacuum Catalyzation (QVC) program that is *actually* integrated, as distinct from the parent Topological Ginzburg–Landau (TQGL) functional written in the companion preprint. The time stepper is a truncated two-dimensional Gross–Pitaevskii equation (GPE) coupled to a first-order dynamical Casimir (DCE) amplitude. Chern–Simons, Berry, and Skyrme terms are omitted from the equation of motion. The gap is $\Delta_{\text{live}}[\psi]$. Over 1 ps the live gap rises from 0.6000 meV to 0.6305 meV; it does not collapse to 0.032 meV. Fold-slaved DCE over 80 ps saturates at $A^*/A_c^{\text{fold}} = 0.9117$ (resonant) and 0.8800 (detuned), not 0.52. Static comparison remains the Option A* fold at $\lambda^* = 0.7628$, $\Delta^* = 0.2324$ meV.

Contents

1 Scope and relation to the preprint

The companion preprint freezes Option A* (fat torus, Tier-C vacuum form, derived α - E_{vac} bridge) and proves that the static gap map is a saddle-node fold. That paper also writes a parent functional

$$\mathcal{F}[\Psi, \mathbf{a}] = \mathcal{F}_{\text{bulk}} + \mathcal{F}_{\text{CS}} + \mathcal{F}_{\text{Berry}} + \mathcal{F}_{\text{ZPF}} + \mathcal{F}_{\text{Skyrme}}. \quad (1)$$

The present document answers a narrower question: which pieces of $\mathcal{F}$ enter a time-dependent integrator, how the gap is read from the field, and how G_{ij} is presently computed. We do not derive TDVT spacetime, and we do not upgrade Wigner–Seitz overlaps to a torsion vertex (hypothesis H-TOR1 remains open).

Numbers used without further comment are the locked macros: $E_{\text{vac}} = 0.1287 \text{ meV}$, $F = 0.58068$, $\alpha_c = 0.1818$, $\lambda_{\text{fold}} = 0.8475$, $\lambda^* = 0.7628$, $\Delta_0 = 0.600 \text{ meV}$, $\Delta_c = 0.109 \text{ meV}$, $\Delta^* = 0.2324 \text{ meV}$, $Q_H = 1/N$, $\hbar c_s = 0.07 \text{ meV} \cdot \mu\text{m}$.

2 What is in the equation of motion

In the EOM	Not in the EOM
2D cubic–quintic GPE	Chern–Simons $\mathbf{a} \wedge d\mathbf{a}$
toroidal $V_{\text{ZPF}} \propto E_{\text{vac}}$	Berry $\Omega(\mathbf{k}) \cdot \mathbf{j}$
coupled DCE amplitude ODE	Skyrme $ \nabla^2 \Psi ^2$
live $\Delta = \Delta_0 \sqrt{n_w/n_{\text{ref}}}$	prescribed $\Delta(t)$ stages

Figure 1: Terms retained in the integrator (left) versus terms retained only in the parent functional or withdrawn as a gap schedule (right).

The condensate field $\psi(x, y, t)$ obeys a cubic–quintic GPE in two spatial dimensions,

$$i\hbar\partial_t\psi = \left[-\frac{\hbar^2}{2m^*}\nabla^2 + \alpha_0|\psi|^2 + \beta|\psi|^4 + V_{\text{ZPF}}(\mathbf{r}; A) + V_{\text{ph}}(\mathbf{r}, t) \right] \psi, \quad (2)$$

with $\hbar = 0.6582119569 \text{ meV}\cdot\text{ps}$ in the integrator units. The vacuum potential is localized on the torus ring,

$$V_{\text{ZPF}}(\mathbf{r}; A) = -A E_{\text{vac}} \exp\left(-\frac{(\rho - R)^2}{2r^2}\right), \quad \rho = \sqrt{x^2 + y^2}, \quad (3)$$

and E_{vac} is the locked Tier-C value 0.1287 meV , never 0.244 meV . Explicitly, $V_{\text{ZPF}} = -E_{\text{vac}}(A(t)/A(0))$ times a ring Gaussian. The phonon drive on the 1 ps window is $V_{\text{ph}} = 0.1\Delta_0 \sin(\omega_0 t)$ on the same ring, with $\omega_0 = 2\Delta_0/\hbar$.

The dynamical Casimir amplitude obeys the first-order law actually coded in the suite,

$$\dot{A} = [\Lambda(\Delta) \mathcal{L}(\delta\omega/\Gamma) - \Lambda_{\text{loss}}] A, \quad \Lambda(\Delta) = \frac{m\Delta}{2\hbar} \frac{Q}{2\pi}, \quad \mathcal{L}(x) = \frac{1}{1+x^2}, \quad (4)$$

with $\Lambda_{\text{loss}} = \Gamma_{\text{cav}}(\Delta^*)$ the cavity linewidth, not a parameter fitted to $A^*/A_c = 0.52$. Resonant driving uses $\omega_{\text{pump}}/2 = 2\Delta^*/\hbar$; the detuning-feedback run uses $\omega_{\text{pump}}/2 = 2\Delta^*/\hbar + 2\Gamma$. On the coupled 1 ps window, $\Delta = \Delta_{\text{live}}[\psi]$. On the 80 ps window, Δ is slaved to the physical fold branch $\Delta(A)$. The fold critical amplitude is $A_c^{\text{fold}} = 1.5582 \mu\text{m}^{-1}$, not a BKT value.

Chern–Simons $\mathbf{a} \wedge d\mathbf{a}$, Berry $\Omega(\mathbf{k}) \cdot \mathbf{j}$, and Skyrme $|\nabla^2 \Psi|^2$ do *not* appear in Eq. (??). Any Hall staircase or hopfion charge quoted from dynamics is therefore an interpretation using the parent functional, not a flux measured on the grid.

3 Live gap estimator

A previous solver stored

$$\Delta(t) \in \{0.600, 0.389, 0.200, 0.032\} \text{ meV} \quad (5)$$

as a function of time and did not use ψ . That function is not reimplemented.

Let $w(x, y) = \exp[-(\rho - R)^2/(2r^2)]$ be the ring weight. Define

$$n_w[\psi] = \frac{\sum_{ij} w_{ij} |\psi_{ij}|^2}{\sum_{ij} w_{ij}}. \quad (6)$$

The primary estimator is

$$\Delta_{\text{live}}[\psi] = \Delta_0 \sqrt{\frac{n_w[\psi]}{n_{\text{ref}}}}, \quad n_{\text{ref}} = n_w[\psi(\cdot, 0)]. \quad (7)$$

Equation (??) has no time argument. If $n_w(0) = n_{\text{ref}}$ then $\Delta_{\text{live}}(0) = \Delta_0$. Secondary diagnostics are the density-weighted cubic–quintic potential $\langle \alpha_0 |\psi|^2 + \beta |\psi|^4 \rangle_n$ and the first-order coherence

$$g^{(1)} = \frac{|\langle \psi \rangle|^2}{\langle |\psi|^2 \rangle}. \quad (8)$$

Coherence is a reportable output, not a license to omit Lindblad dissipation. The present truncation is Hamiltonian plus the DCE ODE; laboratory collapse would be weaker.

4 Static comparison (fold, not a schedule)

Every dynamical run is compared to the locked fold, not to withdrawn stages. At working participation,

$$\lambda^* = 0.7628, \quad \alpha^* = 0.90 \alpha_c, \quad \Delta^* = 0.2324 \text{ meV} \quad (9)$$

on the physical branch. The integrator should be initialized near this static point (or near Δ_0 with λ ramped), never at 0.389 meV. The fold theorem forbids a static physical target of 0.032 meV: $\delta = 0.032/0.6 < \delta_c = 0.1818$. Maximum static reduction is 81.8% at $\Delta_c = 0.109 \text{ meV}$.

Drive-scale diagnostics that multiply historical rates by $E_{\text{vac}}/0.244 \approx 0.527$ are not split-step results and are not tabulated as physics.

5 Initial data and grid

The completed run uses a square box of side 200 nm, grid 128×128 , and $dt = 2$ fs for a 1 ps coupled window. Kinetic coefficient $\hbar^2/(2m^*) = \Delta_0 a_H^2$ with $a_H = r = 8$ nm; cubic–quintic parameters $\alpha = -\Delta_0$, $\beta = \Delta_0/n_0$, $n_0 = 1$. The initial field is a hopfion with 10% core depletion, phase $Q_H\theta$ with $Q_H = 1/N = 1/5$, and 3% noise. The reference density n_{ref} is computed once from this field so that $\Delta_{\text{live}}(0) = \Delta_0$. Time stepping is split-step Fourier for Eq. (??) and RK4 for Eq. (??). No output column named `delta_prescribed` exists.

6 Coupled GPE results (1 ps)

Table 1: Live-gap GPE window. All entries use ψ ; none is a time schedule.

Quantity	Value
$\Delta_{\text{live}}(0)$	0.6000 meV
$\Delta_{\text{live}}(1 \text{ ps})$	0.6305 meV
$\min \Delta_{\text{live}}$	0.6000 meV
$\text{live} < \Delta_c$	False
$g^{(1)}(1 \text{ ps})$	0.8832
$A(1 \text{ ps})/A_c^{\text{fold}}$	0.9811
$\ \psi\ $ relative drift	1.30×10^{-14}

The live gap *increases*. That is the result. It is not a 92% collapse, and it is not spun as a delayed catalyzation: the truncated Hamiltonian plus a 1 ps drive simply did not deplete the ring-weighted density below n_{ref} .

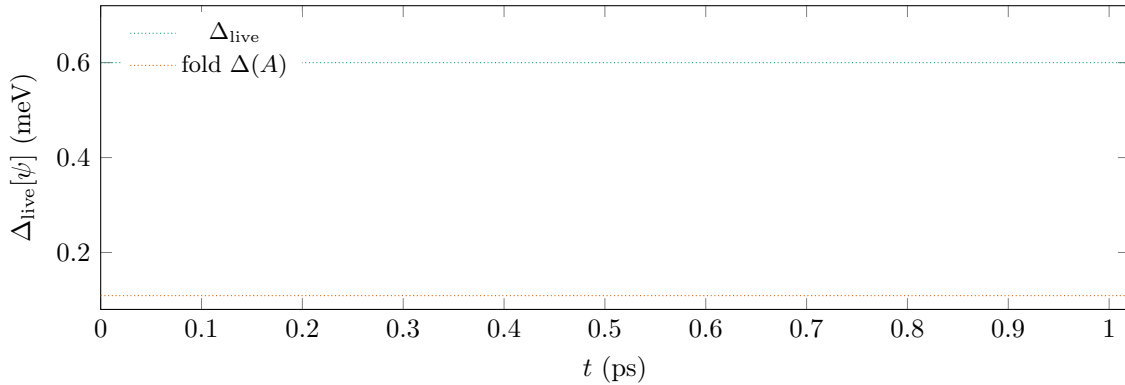


Figure 2: Live gap versus the instantaneous fold branch $\Delta(A)$ on the 1 ps window. Guides: Δ_0 and Δ_c .

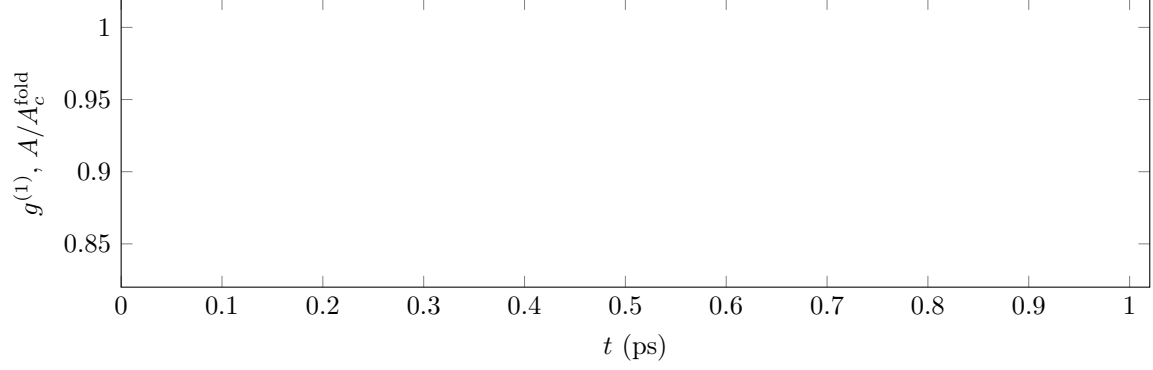


Figure 3: Coherence $g^{(1)}$ and A/A_c^{fold} on the same window.

7 Fold-slaved DCE (80 ps)

Table 2: DCE saturation. Ratios are computed from $A_c^{\text{fold}} = 1.5582 \mu\text{m}^{-1}$; 0.52 is not imposed.

Case	A^*/A_c^{fold}	Δ_{ss} (meV)	$L(0)$
resonant	0.9117	0.2240	1.000
detuning	0.8800	0.2461	0.200

Neither run folded out. Two-timescale justification: Γ_{cav}^{-1} is tens of picoseconds, much longer than the 1 ps GPE window, so the 80 ps amplitude uses Δ slaved to the physical fold branch rather than a live ψ .

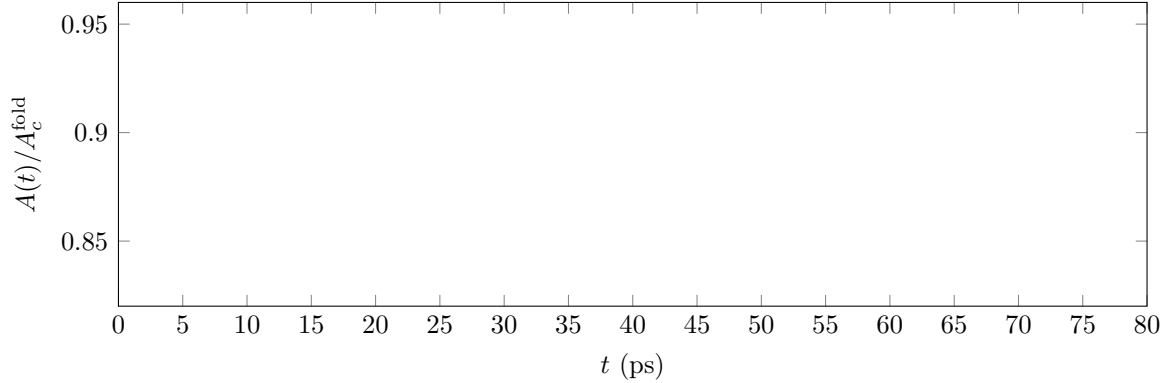


Figure 4: Fold-slaved DCE amplitude in units of A_c^{fold} . Resonant (solid) and detuned (dashed) saturations are 0.9117 and 0.8800.

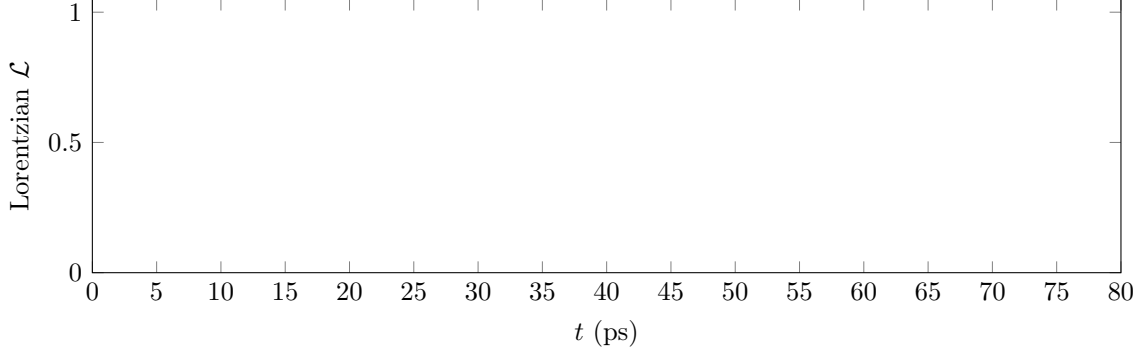


Figure 5: Lorentzian $\mathcal{L}(\delta\omega/\Gamma)$ on the same 80 ps runs.

8 Coupling matrix G_{ij}

8.1 Democratic lock

For $G = g_0(J_N - I_N)$ the spectrum is exact:

$$\lambda_{\max} = (N - 1)g_0 \quad (\text{mult. } 1), \quad \lambda_{\min} = -g_0 \quad (\text{mult. } N - 1). \quad (10)$$

Helmert (sum-zero) vectors span the dark subspace. The catalyzon shift is $\Delta_{\text{cat}} = \Delta_{\text{ex}} - g_0$. Working $g_0 = 0.21$ meV is larger than E_{vac} and is withdrawn as a vacuum scale. The suite identification used in Schwinger/Berry diagnostics is $g_0 = \lambda^* E_{\text{vac}} = 0.0982$ meV, so $|\lambda_{\min}| = 0.0982$ meV. The older scaling $g_0 \approx 0.0527$ meV ($\lambda_{\text{cat}} \approx 0.410$) remains a distinct guess from $0.1 \times E_{\text{vac}}/0.244$ and is not mixed with 0.0982 meV.

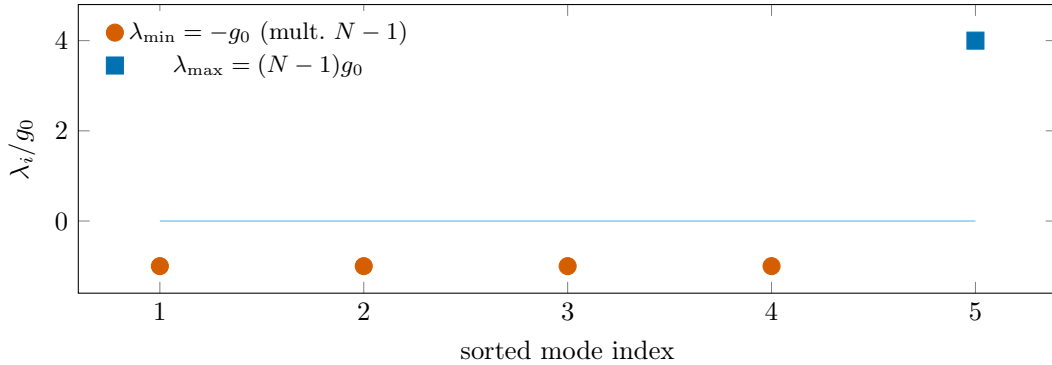


Figure 6: Democratic spectrum used as the algebraic lock. WS numerics are compared to this figure, not to $\lambda_{\min} = -4g_0$.

8.2 Wigner–Seitz program computation

Five Gaussians of width $\sigma \approx 0.18 \lambda_M$ are placed at the vertices of a regular pentagon inscribed in the moiré hexagon of period $\lambda_M = 13$ nm. Off-diagonal entries are overlaps against $K_0(|\mathbf{r} - \mathbf{r}'|/\xi_K)$ with $\xi_K = R = 50$ nm; the diagonal is set to zero. Democratic $J - I$ is not inserted. Reported acceptance tests are: Hermitian, $G_{ii} = 0$, $\text{Tr } G = 0$, $\lambda_{\min} < 0$, and relative Frobenius distance $\approx 7.0\%$ to the democratic matrix at $\langle G_{\text{off}} \rangle$. Dark multiplicity is split, as expected for a pentagon in a hexagon.

Eigenvalues remain $\mathcal{O}(10^{-4})$ in kernel units; converting to meV requires the missing prefactor κ such that $g_0 = \lambda_G E_{\text{vac}}$.

Archival heatmaps `figures/Gij_heatmap.png` and `figures/Gij_spectrum.png` document this run. They are program-level, not a torsion variation of $S_{\text{FN}} + S_{\text{sf}} + S_{\text{int}}$.

8.3 H-TOR1 remains open

The conjectured vertex

$$G_{ij} \stackrel{?}{=} \frac{\kappa_{\text{tor}}}{4\pi^2} \int_{\text{WS}} \varepsilon^\mu{}_{\nu\rho\sigma} \partial_\sigma q_H \Phi_i^{*\nu} \Phi_j^\rho d^3x \quad (11)$$

is not evaluated. Passing WS acceptance tests does not close H-TOR1.

9 Parallel channels (do not stack)

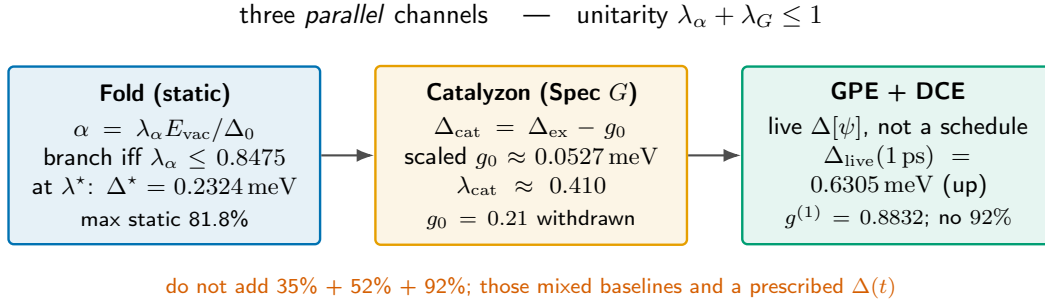


Figure 7: Fold, catalyzon, and GPE/DCE are parallel. The withdrawn 35/52/92 stack mixed baselines and a prescribed $\Delta(t)$.

The completed 1 ps run has $\Delta_{\text{live}}(1 \text{ ps}) = 0.6305 \text{ meV} > \Delta_c$, so it is not an overshoot of the fold. A future run that did fall below Δ_c would be flagged unphysical versus the static branch and would still not resurrect 0.032 meV.

10 Schwinger and Berry (not in the EOM)

With $g_0 = \lambda^* E_{\text{vac}} = 0.0982 \text{ meV}$ one has $E_{\text{pair}} = 2|\lambda_{\text{min}}| = 0.1963 \text{ meV}$, replacing withdrawn 0.172 meV. The diagnostic crossover is $T^* = 118.7 \text{ mK}$. The GOE/GUE Monte Carlo ratio of mean $|\lambda_{\text{min}}|$ is $R_{\text{mean}} = 0.908$, not a forced 1.43 enhancement. These numbers are not terms in Eq. (??).

11 Withdrawn dynamical claims

The v3 “three-stage collapse” identified (i) bare $\rightarrow$ self-consistent $0.600 \rightarrow 0.389 \text{ meV}$ as a 35% static loop at $E_{\text{Cas}} = 0.244 \text{ meV}$; (ii) a 52% catalyzon mix $(\Delta_{\text{QVC}} - 0.1) / \Delta_{\text{ex}}$; (iii) a coherent 92% drop $(0.389 - 0.032) / 0.389$ over 1 ps with $V_{\text{ZPF}} = -0.244 \text{ meV}$. Stage (i) is fold-out under the derived bridge. Stage (ii) is not Spec(G) at $g_0 = 0.0982 \text{ meV}$. Stage (iii) used a gap function of time, not of ψ , and landed below δ_c . The live-gap suite instead gives 0.6305 meV at 1 ps.

12 Reading of the completed run

Figures ??–?? report $\Delta_{\text{live}}(t)$, $g^{(1)}(t)$, and $A(t)$ with guides at Δ_0 and Δ_c . Because Δ_{live} rose while $g^{(1)}$ stayed of order one, the 1 ps window is a coherent non-collapse of the truncated GPE, not catalyzation through the fold and not dephasing. Hall conductance is not an output of Eq. (??); it remains a prediction of $Q_H = 1/N$ awaiting a Chern–Simons integrator.

13 Reproducibility

Freeze and fold numbers are regenerated from `qvc.bridge`, `qvc.gap_fold`, and `qvc.gap_reduction`. Dynamics are in `output/tqgl_v4/results.json` and `csv/`, ingested here as `data/coupled_timeseries.csv` and `data/dce_*.csv`. The live-gap estimator is `qvc.tqgl.delta_live_from_psi`. Compilation of this PDF does not re-execute the GPE.